



Cardiovascular disease detection using a novel stack-based ensemble classifier with aggregation layer, DOWA operator, and feature transformation

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ABSTRACT

Due to their widespread prevalence and impact on quality of life, cardiovascular diseases (CVD) pose a considerable global health burden. Early detection and intervention can reduce the incidence, severity, and progression of CVD and prevent premature death. The application of machine learning (ML) techniques to early CVD detection is therefore a valuable approach. In this paper, A stack-based ensemble classifier with an aggregation layer and the dependent ordered weighted averaging (DOWA) operator is proposed for detecting cardiovascular diseases. We propose transforming features using the Johnson transformation technique and normalizing feature distributions. Three diverse first-level classifiers are selected based on their accuracy, and predictions are combined using the aggregation layer and DOWA. A linear support vector machine (SVM) meta-classifier makes the final classification. Adding the aggregation layer to the stacking classifier improves classification accuracy significantly, according to the study. The accuracy is enhanced by 5%, resulting in an impressive overall accuracy of 94.05%. Moreover, the proposed system significantly increases the area under the receiver operating characteristic (ROC) curve compared to recent studies, reaching 97.14%. It further reinforces the classifier's reliability and effectiveness in classifying cardiovascular disease by distinguishing between positive and negative instances. With improved accuracy and a high area under the curve (AUC), the proposed classifier exhibits robustness and superior performance in the detection of cardiovascular diseases.

1. Introduction

Cardiovascular disease (CVD) detection is a critical focus within healthcare due to its status as the leading cause of death globally. A substantial burden is placed on healthcare systems by these diseases, including coronary artery disease, heart failure, and stroke [1]. According to the World Health Organization report, in 2019, about 17.9 million people died due to CVDs, which includes 32% of all global deaths. Also, out of 17 million premature deaths (under 70 years old) due to non-communicable diseases, 38% were caused by CVDs [2]. The most important behavioral risk factors for cardiovascular disease include unhealthy diet, inactivity, smoking, and alcohol consumption [2,3]. The effects of these risk factors may appear in people in the form of increased blood pressure, blood glucose, blood lipids, and overweight and obesity [2–6].

Therefore, Prompt and accurate detection of CVDs can lead to early intervention, significantly improving the prognosis for individuals with

such conditions [7]. The goal of CVD detection is to identify heart and blood vessel anomalies early on, allow for timely management and treatment, and ultimately, reduce mortality and improve quality of life [8]. Effective detection strategies can also minimize the economic burden of CVDs through the prevention of costly complications, thus reducing the overall healthcare expenditure associated with chronic management of late-stage heart disease [9]. In research studies, various approaches and methodologies were explored to develop advanced detection models for CVDs.

Machine learning (ML) methods were extensively used to detect CVDs, using algorithms to extract patterns from data [10–12]. The effectiveness of traditional ML algorithms such as logistic regression (LR), decision trees (DTs), support vector machines (SVMs), random forest (RF), K-nearest neighbors (KNN), naïve bayes (NB), and multilayer perceptron (MLP) for predicting CVD risk was investigated [13–15]. In

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addition, ensemble learning methods, such as RFs and gradient boosting, were used to enhance predictive accuracy by combining multiple models [16–19]. Detecting CVD using these approaches demonstrated promising results, underscoring their potential for clinical use.

Due to its ability to automatically extract intricate patterns from complex data, deep learning (DL), a subset of ML, has gained attention in CVD detection. In this study, convolutional neural networks (CNNs) were used to analyze electrocardiograms (ECGs), medical images, and electronic health records to predict CVD outcomes [20–22]. Models based on DL can capture subtle features and improve CVD detection accuracy.

A recent study incorporated multi-omics data, including genomics, proteomics, metabolomics, and transcriptomics, to enhance CVD prediction models [23]. These approaches combined information from different molecular levels to identify novel biomarkers and molecular pathways associated with CVD. It was demonstrated that the integration of multi-omics data can improve the accuracy and precision of CVD models, providing a more comprehensive understanding of the underlying biological mechanisms [24].

The use of explainable AI (XAI) techniques in CVD detection is in line with the growing emphasis on transparency and interpretability in healthcare applications. To provide interpretable explanations for predictions and support clinical decision-making, models such as decision trees, rule-based systems, and Bayesian networks were employed [25, 26]. By highlighting key features and risk factors for CVD outcomes, these approaches enhanced trust and understanding among healthcare professionals.

CVD detection gained traction as a result of personalized risk assessment, a concept that aims to tailor preventative strategies based on an individual's health profile and characteristics [27]. ML algorithms were combined with patient-specific data, including demographics, clinical history, and lifestyle factors, to develop personalized risk prediction models. Health providers could use such models to identify high-risk individuals and take proactive measures to prevent CVDs [28,29].

While CVD detection has advanced, challenges such as handling large datasets, ensuring model interpretability, achieving accurate predictions, and generalizing across diverse populations persist [30].

This study introduces the dependent ordered weighted averaging (DOWA) operator and incorporates it as an aggregation layer in a stacking classifier, presenting a new classifier. In this new stacking classifier, first-level classifiers are selected based on two criteria: accuracy and diversity. The classifier chosen as the meta-classifier is the one that demonstrates the best performance in detecting CVD. Additionally, raw data undergoes a better representation through feature transformation techniques and is used to feed the introduced classifier. The main contributions of this paper can be summarized as follows.

- By introducing a DOWA operator, we propose a new version of the stacking classifier.
- Feature transformation techniques enhance feature representation and quality in our model. We perform comprehensive hyperparameter tuning for maximum accuracy.
- Using the two criteria of accuracy and diversity, the best classifiers are selected to be used as first-level classifiers and meta-classifier.
- Our proposed model achieve an impressive 94.05% accuracy and 97.14% area under the curve (AUC) for cardiovascular diseases. Consequently, a more robust and reliable model was innovated.

The rest of this article is structured as follows. Section 2 introduces related work on predicting cardiovascular diseases using the stacking classifier. Section 3 provides an overview of the comprehensive exposition of the employed dataset and the methodologies. The simulation results and accompanying discussion are presented in Section 4. Section 5 discusses future research directions. Finally, Section 6 draws the conclusions.

2. Related work

The purpose of this section is to discuss several research works and the different methodologies that other researchers have used to approach the problem. The number of patients with cardiovascular disease has increased exponentially [31]. To predict the occurrence of cardiovascular disease, scientists use a variety of methodologies and algorithms. Cardiovascular disease prognosis has been the subject of many research studies over the years. Studies related to this topic will be discussed in the following paragraphs.

An ensemble classifier was used to predict cardiovascular disease early in [32]. ExtraTrees classifier, RF, extreme gradient boosting (XGBoost), and others were incorporated into the proposed framework. According to the authors, the ensemble classifier outperformed individual classifiers on the IEEE Data Port dataset, achieving a 2% increase in accuracy. In [33], the authors presented a design for building an Intelligent Hybrid Framework using different algorithms and 10-fold cross-validation methods on both complete and chosen characteristics. Four different algorithms were used to select features, including LASSO and Relief algorithms. The authors calculated Chi-Square and *P*-value after implementing feature selection techniques.

With the technique of majority voting ensemble in [34], various algorithms were utilized, resulting in an accuracy rate of 90%. They conducted a majority vote of all algorithms in order to improve their overall accuracy. As part of the investigation, the authors computed correlations with the target and examined the relationship between other attributes and positive data points. Using a Decision Tree-like structure, the authors in [35] performed data splits at multiple levels using the chi-squared automated interaction detector technique (CHAID). Using the CHAID decision tree, cardiologists can gain a deeper understanding of a patient's health status, enhancing their ability to distinguish diseases. The dataset was analyzed using CHAID to improve understanding. To enhance the overall accuracy, a majority voting process was implemented. MLP, LR, and NB were used for the voting.

Alghahtani et al. in [36], the stacking classifier structure Was utilized twice. Initially, a stack model was implemented using DT, KNN, and XGBoost classifiers as first-level classifiers, with a RF serving as the meta-classifier. The second stack model comprises deep neural network (DNN), keras deep neural network (KDNN), and the first stacking model as first-level classifiers, utilizing majority voting as the meta-classifier. Their experimental results indicated that the ML ensemble model forecasts the disease most accurately, with 88.70% accuracy.

In [37], the authors presented an ensemble framework using ten different classifiers, including SVM, KNN, LR, and so on. To avoid overfitting, the ensemble framework used the LR as a meta learner to select the best base learners. A fusion of Heart Datasets from various UCI ML repositories and a publicly available Heart Attack Dataset were used to validate the proposed algorithm. Experimental results showed that the stacking classifier performed significantly better than ten single classifier models in terms of accuracy and applicability.

The purpose of this research is to present a new version of the stacking classifier using a novel layer. Furthermore, the effectiveness of the proposed classifier is determined by the careful selection of first and second-level classifiers.

3. Methodology

For the purpose of predicting cardiovascular diseases, this section provides details about the dataset. In Section 3.3, our focus shifts to introducing a novel classifier specifically designed for cardiovascular disease prediction. A meticulous examination of the selection of single classifiers within the stacking classifier is carried out. Through the aggregation layer, we also introduce the DOWA operator and explain how it is integrated into the stacking classifier.

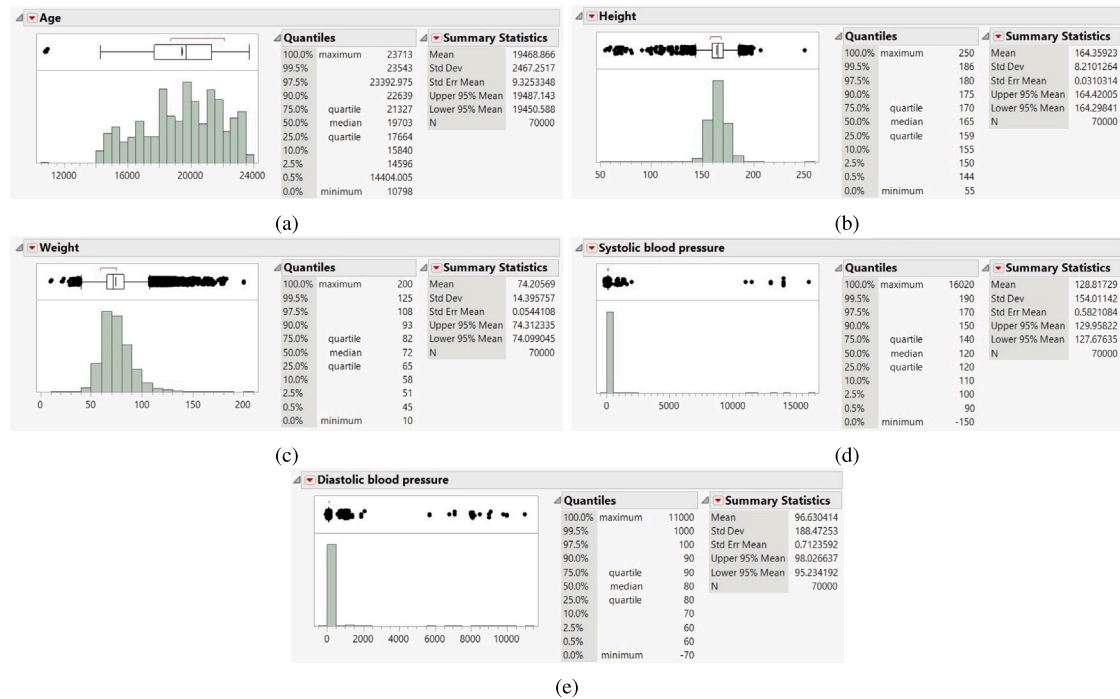


Fig. 1. The statistical characteristics of the continuous features of the dataset, (a) Age, (b) Height, (c) Weight, (d) Systolic blood pressure, (e) Diastolic blood pressure.

3.1. Dataset

For the ML task in cardiovascular diseases, we utilize a Kaggle dataset [38]². Each patient is represented by 11 features in this dataset, which consists of information from 70,000 patients. In the provided dataset, three types of input features are present including; (i) “Objective” features which include factual information such as age (in days), height (in cm), weight (in kg), and gender (1, 2). (ii) “Examination” features which encompass medical examination results like systolic and diastolic blood pressure, cholesterol (1: normal, 2: above normal, 3: well above normal), and glucose levels (1: normal, 2: above normal, 3: well above normal). And (iii) “Subjective” features involving patient-provided information like smoking (0, 1), alcohol intake (0, 1), and physical activity (0, 1). The dataset’s target variable is the presence or absence of cardiovascular disease, categorized as binary. All the values in the dataset are collected during the patients’ medical examinations.

Fig. 1 provides statistical information about the continuous features in the dataset, including weight, diastolic blood pressure, systolic blood pressure, and others. On the other hand, Fig. 2 displays statistical information about the discrete features such as cholesterol, glucose, physical activity, and others.

3.2. Feature transformation

It involves applying mathematical or statistical operations to the features of a dataset to improve the representation and performance of ML models. In our study, we employ the Johnson transformation to normalize the data and mitigate potential biases introduced by features with larger values. In the Johnson transformation, we apply a function to the data to achieve a more symmetric and Gaussian distribution. Specifically, we apply it to all continuous features.

Fig. 3 provides a visualization of the feature transformation process specifically for the “weight” feature. By utilizing (1), we achieve a transformation from the probability plot of the original data to the

probability plot of the transformed data, as depicted in Fig. 3. This process enhances the data’s distributional properties and contributes to improved model performance.

$$TF = -1.41815 + 2.10448 \times \frac{A \sinh(X - 57.3035)}{21.2098}. \quad (1)$$

3.3. Proposed CVD system: Architecture, design, and analysis

The proposed cardiovascular disease prediction system is described in this section in terms of its architecture, design, and analysis. As a first step, we examine the stacking classifier, followed by the dependent ordered weighted averaging operator. The DOWA operator is used to integrate the aggregation layer into the stacking classifier.

3.3.1. Stacking classifier

The stacking classifier, a popular ensemble learning technique, combines the predictions of multi base classifiers to improve prediction accuracy and robustness [39]. In our proposed system, we carefully select a set of base classifiers with diverse characteristics to capture different aspects of the complex relationships present in CVD prediction. As illustrated in Fig. 4, it operates in two stages including:

• Training Stage:

- Multi base classifiers, denoted as C_1 , C_2 , and C_3 , are trained on the transformed dataset.
- Each base classifier learns from the input features and corresponding labels to generate diverse predictions.
- The training process allows the base classifiers to capture different patterns and relationships in the dataset.

• Prediction Stage:

- After training, the base classifiers are used to make predictions on new, unseen data.
- Each base classifier independently provides a prediction (P_1 , P_2 , P_3) for the test data based on its learned knowledge.

² <https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset>

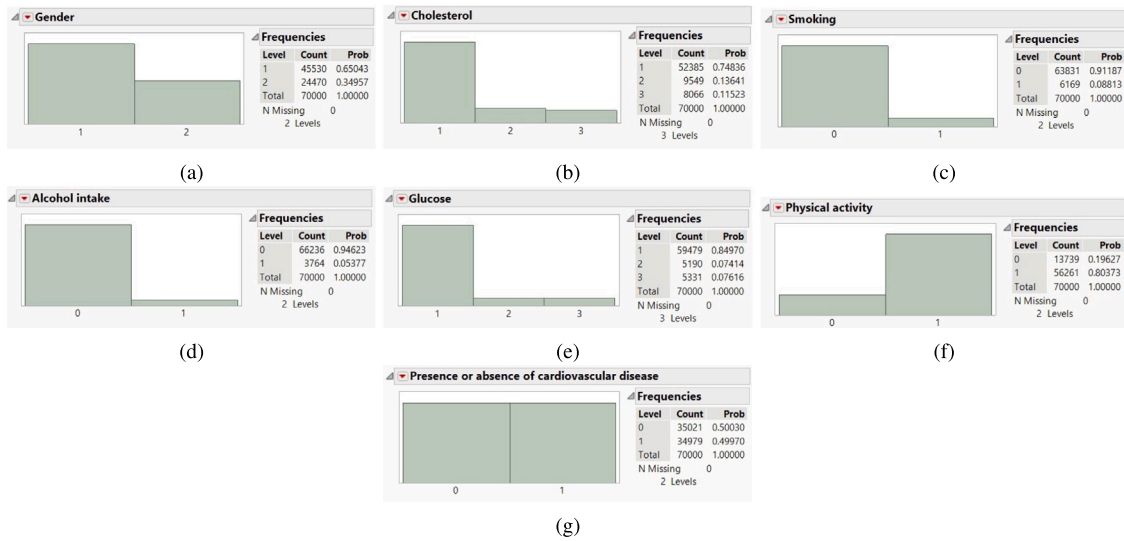


Fig. 2. The statistical characteristics of the discrete features of the dataset, along with the target variable, (a) Gender, (b) Cholesterol, (c) Smoking, (d) Alcohol intake, (e) Glucose, (f) Physical activity, (g) Presence or absence of CVD.

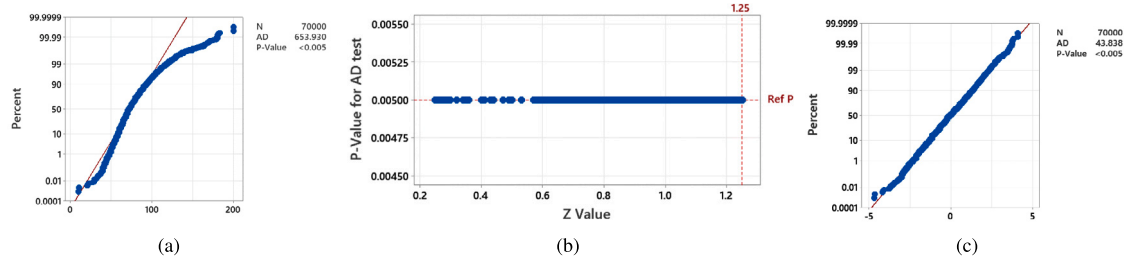


Fig. 3. Feature transformation using the Johnson transformation for the 'Weight' feature from the dataset, (a) Probability plot for original data, (b) Select a transformation, (c) Probability plot for transformed data.

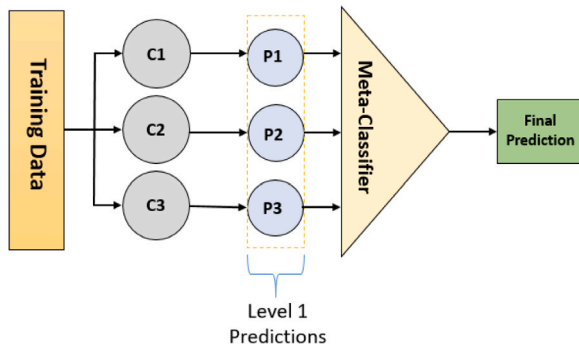


Fig. 4. The stacking classifier employing three first-level classifiers $C1, C2, C3$ for initial predictions, and a meta-classifier for the final predictions.

- These predictions serve as input to the meta-classifier, which combines and aggregates them to generate the final prediction.

By leveraging the collective wisdom of the base classifiers, the stacking classifier improves prediction accuracy and robustness. With multiple classifiers, each with its own strengths and weaknesses, a stacking classifier can capture a greater range of information and make more accurate predictions.

3.3.2. Classifier selection

To achieve optimal prediction performance, first-level classifiers and meta-classifiers need to be selected carefully. In our approach,

we consider both accuracy and diversity metrics when choosing the first-level classifiers, namely $C1, C2$, and $C3$. The classifier with the highest correct classification rate on the test data is chosen as the meta-classifier.

3.3.2.1. First-level classifier selection. Our initial priority is to select classifiers that are highly accurate individually. According to Table 1, we can assess each classifier's performance based on its accuracy on the test data. We aim to incorporate classifiers that provide distinct predictions for different samples, however, recognizing the importance of diversity in ensemble learning. The diversity of perspectives ensures a broader range of perspectives and enhances the system's predictive capability.

The concept of correlation among classifiers is used to evaluate diversity. In order to increase the likelihood of obtaining complementary predictions, we select classifiers that do not exhibit high correlation. With this selection process, the ensemble is able to capture a wider range of patterns and improve prediction accuracy. During the prediction generation of the first-level classifiers, we employ the K-Fold validation method to mitigate potential data leakage issues. Overfitting is prevented by this technique, which ensures robustness.

Based on their exceptional accuracy in predicting cardiovascular diseases, we carefully select three classifiers, namely decision tree, random forest, and gradient boosting. These classifiers are chosen for their ability to capture complex relationships and exhibit strong predictive power. Additionally, we assess their diversity by examining the correlation matrix, as depicted in Fig. 5. Classifiers based on decision trees, random forests, and gradient boosting are correlated visually in the correlation matrix. In the matrix, each cell represents the correlation coefficient between two classifiers. A matrix's color intensity indicates

Table 1

The accuracy of the individual classifiers on the test data.

Classifier	Accuracy
Logistic Regression	70.20 ± 0.28%
Support Vector Machine	59.33 ± 0.33%
k-Nearest Neighbors	68.46 ± 0.21%
Gradient Boosting	73.75 ± 0.19%
Stochastic Gradient Descent	69.52 ± 0.25%
Decision Tree	72.65 ± 0.24%
Random Forest	73.12 ± 0.18%
Multi-Layer Perceptron	66.42 ± 0.23%
Adaptive Boosting	71.87 ± 0.20%
Hard Voting	70.81 ± 0.31%

Table 2

Performance metrics for selecting the meta-classifier.

Classifier	Accuracy	AUC
Logistic Regression	92.87 ± 0.19%	95.33 ± 0.13%
Linear SVM	94.05 ± 0.19%	97.14 ± 0.12%
k-Nearest Neighbors	92.66 ± 0.25%	94.63 ± 0.15%
Gradient Boosting	93.94 ± 0.20%	95.22 ± 0.10%
Decision Tree	90.48 ± 0.29%	90.69 ± 0.27%
Random Forest	90.67 ± 0.26%	95.32 ± 0.13%
Multi-Layer Perceptron	94.03 ± 0.19%	97.14 ± 0.12%
Adaptive Boosting	92.61 ± 0.15%	96.75 ± 0.12%

the level of correlation, with darker colors indicating lower correlation and lighter colors indicating higher correlation. According to this analysis, the selected classifiers displayed relatively diverse prediction patterns, making them suitable candidates for ensemble learning.

Currently, researchers are working on determining the optimal number of classifiers for fusion. A stepwise selection process is used in our approach. By starting with the two classifiers that achieve the highest individual accuracy, we progressively add more classifiers. The process continues as long as adding additional classifiers improves the overall accuracy of the system. As a result of this iterative approach, we determine that three classifiers are optimal for our ensemble: decision trees, random forests, and gradient boosting. Besides exhibiting high individual accuracy, these classifiers also demonstrate a wide range of predictions, enhancing the performance of the ensemble.

3.3.2.2. Meta-classifier selection. Various criteria exist for the selection of the meta-classifier. In this study, we employ two primary criteria, namely accuracy and AUC (an indicator of model robustness), as depicted in Table 2. Among various classifiers, the linear SVM is identified as the optimal choice based on these criteria. Not only does the linear SVM exhibit superior performance, but it also boasts better interpretability compared to its competitive models. A linear SVM identifies the optimal hyperplane that effectively separates the different classes within the input feature space by identifying the optimal hyperplane. It is defined by a linear decision boundary that maximizes the margin, which is the distance between the hyperplane and the nearest data points from each class. It is associated with better generalization and performance of unseen data when there is a larger margin.

The training process of a linear SVM involves finding the parameters that define the hyperplane. This is accomplished by solving a constraint optimization problem, as shown in (2), (3), (4). The objective is to minimize classification errors while simultaneously maximizing the margin. The optimization process determines the optimal set of weights (coefficients) that can best separate the classes while satisfying the margin constraint. Fig. 6 illustrates the operational principles of the linear support vector machine classifier.

$$\text{maximize}_{w,b} \frac{2}{\|w\|}, \quad (2)$$

$$\text{s.t.: } (w^T x_i + b) \geq 1, \quad \text{for } y_i = 1, \quad i = 1, 2, \dots, N. \quad (3)$$

$$(w^T x_i + b) \leq -1, \quad \text{for } y_i = -1, \quad i = 1, 2, \dots, N. \quad (4)$$

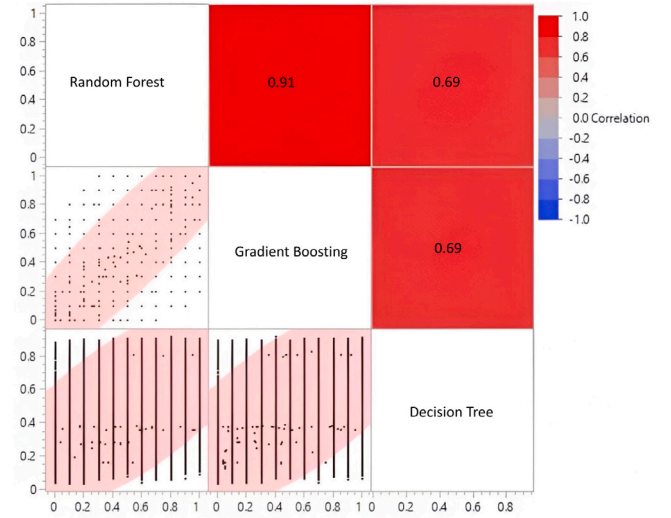


Fig. 5. The correlation between the predictions of the first-level classifiers in the stacking classifier, which serves as a measure of diversity among the classifiers.

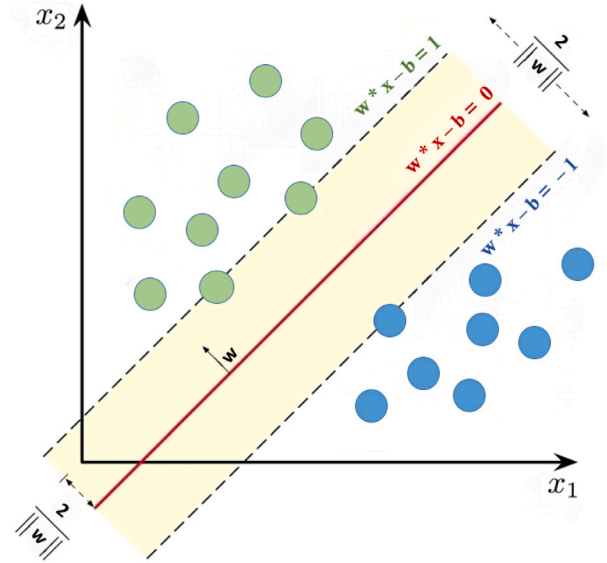


Fig. 6. SVM classifier operates by finding a hyperplane that minimizes classification errors and maximizes the margin between the two classes.

where y_i represents the i th class and N represents the number of samples. By employing the linear SVM as the meta-classifier in our proposed CVD prediction system, we aim to leverage its ability to effectively classifying and separating different classes based on the learned decision boundary.

3.3.3. Aggregation layer

By introducing an aggregation layer, which employs dependent ordered weighted averaging, we extend the capabilities of the stacking classifier. By combining the predictions from the first-level stacking classifiers, we can further refine and enhance the overall predictive ability. In this operator, the soft outputs, which represent the confidence of each sample belonging to a specific class, are combined into a single prediction. The functioning of the DOWA operator is described in Definition 1 [40].

Definition 1. The OWA operator is a mathematical operator that combines inputs based on their weights and order. Decision-making and

aggregation processes commonly use it. The OWA operator aggregates inputs by arranging them in a specific order and assigning weights accordingly. Let $P_1, P_2, \dots, P_n$ be the soft outputs or predictions from the first-level classifiers for a given sample. These predictions represent the probabilities of the sample belonging to each class. and let μ be the average value of these predictions for each class, i.e., $\mu = \frac{1}{n} \sum_{j=1}^n P_j$, $(\beta(1), \beta(2), \dots, \beta(n))$ is a permutation of $(1, 2, \dots, n)$ such that $P_{\beta(j-1)} \geq P_{\beta(j)}$ for all $j = 2, \dots, n$, then we call

$$s(P_{\beta(j)}, \mu) = 1 - \frac{|P_{\beta(j)} - \mu|}{\sum_{j=1}^n |P_j - \mu|}, \quad j = 1, 2, \dots, n, \quad (5)$$

the similarity degree of the j th largest argument $P_{\beta(j)}$ and the average value μ .

The DOWA operator computes the weighted average of the predictions based on a weight vector where each weight represents the importance or significance of the corresponding prediction. Let the $w = (w_1, w_2, \dots, w_n)^T$ be the weight vector of the OWA operator, then we define the following:

$$w_j = \frac{s(P_{\beta(j)}, \mu)}{\sum_{j=1}^n s(P_{\beta(j)}, \mu)}, \quad j = 1, 2, \dots, n, \quad (6)$$

where $s(P_{\beta(j)}, \mu)$ is defined by (5). Clearly, we have $w_j \in [0, 1]$ and $\sum_{j=1}^n w_j = 1$. Since

$$\sum_{j=1}^n s(P_{\beta(j)}, \mu) = \sum_{j=1}^n s(P_j, \mu), \quad (7)$$

then (6) can be rewritten as

$$w_j = \frac{s(P_{\beta(j)}, \mu)}{\sum_{j=1}^n s(P_j, \mu)}, \quad j = 1, 2, \dots, n. \quad (8)$$

In this case, we have

$$OWA(P_1, P_2, \dots, P_n) = \frac{\sum_{j=1}^n s(P_{\beta(j)}, \mu) P_{\beta(j)}}{\sum_{j=1}^n s(P_j, \mu)}. \quad (9)$$

According to (4), we can write

$$OWA(P_1, P_2, \dots, P_n) = \frac{\sum_{j=1}^n s(P_j, \mu) b_j}{\sum_{j=1}^n s(P_j, \mu)}. \quad (10)$$

Depending on the characteristics of the dataset and the first-level classifier, the weights assigned to each prediction can be adjusted. Weights can be optimized to maximize prediction accuracy or align with domain-specific requirements. A meta-classifier makes the final prediction based on the aggregated predictions from the aggregation layer. To predict the target variable, the meta-classifier uses its own learning algorithm to combine the aggregated predictions.

By utilizing the DOWA operator in aggregation, our proposed CVD prediction system offers several advantages. Having the ability to assign higher weights to predictions that are closer to the average prediction of all classifiers is a key advantage. Outlier classifiers that produce significantly different predictions from the majority are mitigated by this feature. The DOWA operator ensures a more balanced and robust final prediction by assigning higher weights to classifiers that exhibit closer alignment with the collective knowledge of all classifiers. A consensus-based approach reduces the potential bias introduced by individual classifiers by considering the collective wisdom and agreement among the base classifiers.

To calculate the weights for each classifier, we employ (8), which considers the predictions made by the first-level classifiers in the stacking classifier. With the weights determined, we aggregate the predictions of all the first-level classifiers for each class using (10). This aggregation process, facilitated by the DOWA operator, combines the individual predictions into a unified prediction that encapsulates the collective knowledge and expertise of the ensemble. After adding the aggregation layer to the stacking classifier, the resulting classifier is shown in Fig. 7. The new classifier demonstrates improved and more reliable performance.

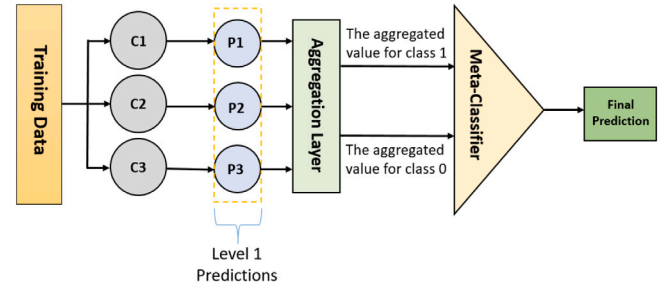


Fig. 7. The stacking classifier after incorporating the aggregation layer into its structure.

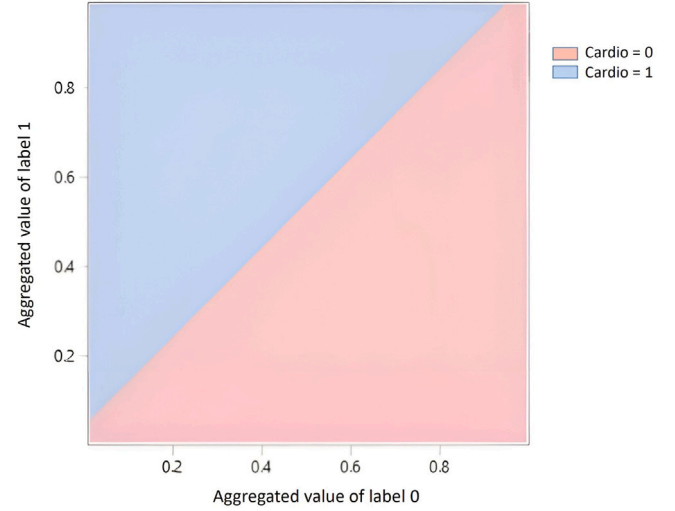


Fig. 8. Decision boundary determined by the linear support vector machine classifier for the two classes.

4. Experiment results and discussion

In this section, we expound upon the outcomes and introduce various performance metrics used to appraise the algorithm's efficacy, including accuracy, sensitivity, precision, F1-Score, specificity, and AUC.

4.1. Performance metrics

4.1.1. Precision

Precision measures the proportion of true positive predictions among all positive predictions made by a model. It focuses on the accuracy of positive predictions, e.g.: $\text{Precision} = \frac{TP}{TP+FP}$.

4.1.2. Specificity

Specificity evaluates the proportion of true negative predictions among all negative predictions made by a model. It emphasizes the accuracy of negative predictions, e.g., $\text{Specificity} = \frac{TN}{TN+FP}$.

4.1.3. Accuracy

Accuracy gauges the overall correctness of a model's predictions by considering both true positives and true negatives relative to the total number of predictions, e.g., $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$.

4.1.4. Recall

"Recall", also known as "sensitivity" or "true positive rate", measures the proportion of actual positive instances that were correctly identified by a model. It focuses on the model's ability to capture all positive instances, e.g., $\text{Recall} = \frac{TP}{TP+FN}$.

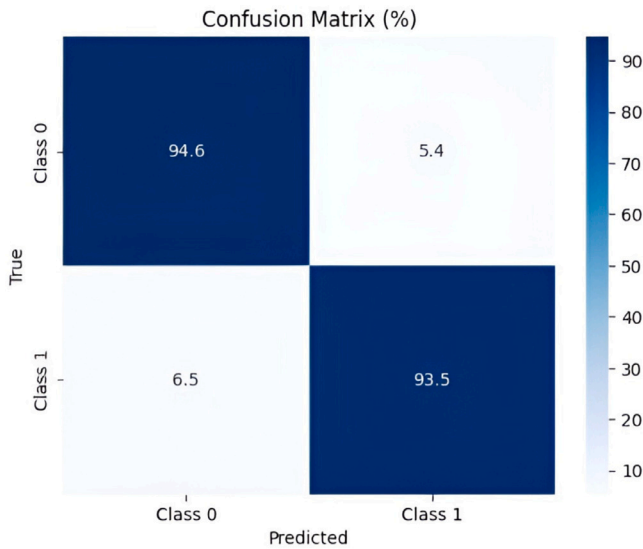


Fig. 9. The confusion matrix for the proposed stack-based ensemble classifier, which achieved an accuracy of 94.05% on the test data.

Table 3
Performance metrics of the proposed method.

Metric	Value
Accuracy	94.05 ± 0.19%
Precision	94.54 ± 0.15%
Recall	93.50 ± 0.17%
Specificity	94.60 ± 0.12%
F1-Score	94.02 ± 0.14%
AUC	97.14 ± 0.12%

4.1.5. F1-Score

F1-score is a balanced metric that considers both precision and recall. It provides a single value that balances the trade-off between precision and recall, making it useful when dealing with imbalanced datasets, e.g., $F1\text{-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$, where TP is true positive, FP is false positive, TN is true negative and FN is false negative.

After incorporating the aggregation layer into the stacking classifier using the DOWA operator and selecting the first-level classifiers and meta-classifier, we evaluate the performance of the meta-classifier on the test data using the K-Fold validation method with $K = 15$. The linear support vector machine classifier is selected as the meta-classifier to distinguish between the two classes, as shown in Fig. 8. The meta-classifier achieves an accuracy of 94.05% on the test data.

The confusion matrix of the meta-classifier on the test data is presented in Fig. 9, indicating that 5.4% of samples from class '0' and 6.5% of samples from class '1' are misclassified. The receiver operating characteristic curve, which illustrates the relationship between the true positive rate (TPR) and the false positive rate (FPR) at various classification thresholds, is depicted in Fig. 10. The ROC curves of both classes exhibit a considerable overlap, indicating a similar performance between the two classes. The area under the ROC curve (AUC) serves as a metric for evaluating the classifier's performance, with a higher AUC value, closer to 1, indicating better classifier performance. The proposed system exhibits an AUC of 97.14%. Other performance metrics of the proposed system are depicted in Table 3.

Based on the results, the linear SVM meta-classifier achieves high accuracy on the test data and exhibits balanced performance in classifying both classes. ROC curve overlap and low misclassification rates indicate the effectiveness of the meta-classifier in predicting cardiovascular diseases.

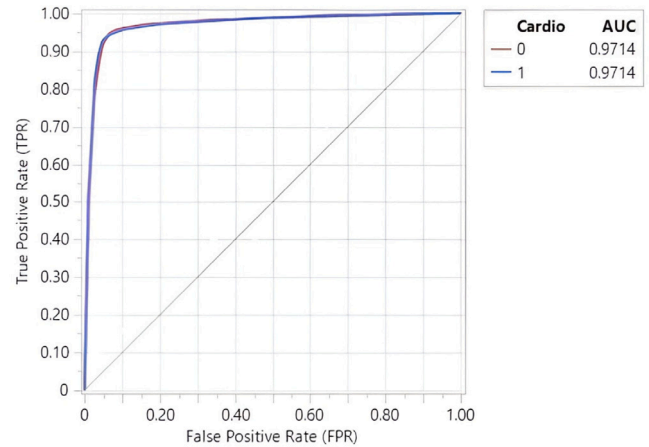


Fig. 10. ROC curve illustrating the classification performance of the proposed model. The area under the curve (AUC) is 97.14%, indicating excellent discriminative power and predictive accuracy.

5. Future research directions

There are several challenges associated with the newly proposed model that future research can address.

- As mentioned, in selecting the optimal number of first-level classifiers, we employed the stepwise selection method. The first challenge could involve exploring alternative approaches for selecting the optimal number of first-level classifiers, such as random search, grid search, and others.
- Another challenge could involve exploring the use of alternative attention-based approaches or leveraging other extensions of the ordered weighted averaging (OWA) operator to implement the aggregation layer.
- The final challenge could involve exploring a deep stacking model. This entails incorporating multiple layers of prediction and aggregation within the stacking classifier structure, potentially leading to a more sophisticated model.
- Given that a historical dataset was used in this research, it is recommended that a cohort study be designed to test the stated methodology on a longitudinal dataset and to evaluate its performance.

6. Conclusion

The rapid and timely detection of cardiovascular diseases is of special importance. Therefore, in this study, by adding an aggregation layer to the structure of the stacking classifier, a new classification method was devised. In the proposed classifier, the aggregation layer was implemented using the DOWA operator, and the first-level classifiers and meta-classifier were carefully selected based on the two criteria of accuracy and diversity. Additionally, raw data from potential CVD patients, after appropriate preprocessing and using feature transformation techniques, were utilized to feed the proposed classifier. The proposed classifier demonstrated a significant increase in accuracy and AUC in the detection of CVD. Furthermore, as it is based on the aggregation of opinions from multiple base classifiers, the proposed classifier is more reliable.

CRedit authorship contribution statement

Mehdi Hosseini Chagahi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Saeed Mohammadi Dashtaki:** Writing –

original draft, Software, Resources, Methodology, Project administration, Funding acquisition, Data curation, Conceptualization. **Behzad Moshiri:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **M.d. Jalil Piran:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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